
Study on the effect of cyclic catalytic pyrolysis on sludge pyrolysis products

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Abstract: Pyrolysis technology has become a hot issue in the industry of harmless treatment of sludge in recent years. The present ectopic catalytic pyrolysis process suffers from the defects of unidirectional flow of pyrolysis steam, poor heating value of the output heavy tar, which is difficult to be reutilized, etc. In this paper, we design a cyclic catalytic cracking by pyrolytic steam process to optimize the above defects: under the optimal experimental conditions (pyrolysis starting temperature was 500 °C, final temperature was 800 °C, iron-based carbon catalyst dosage was 4 g, and motor speed was 1050 r/min), this process generated more pyrolysis gas and light tar. The cyclic catalytic cracking by pyrolytic steam process could increase the residence time of pyrolysis steam in the reaction unit and promoted the secondary cracking, catalytic and reforming effects of tar.

Keywords: Sludge; Pyrolysis; Cyclic catalytic cracking; Secondary cracking; Tar

1. Introduction

Sewage sludge is the solid residue from wastewater treatment, and nowadays, the sludge production is increasing. In China, sludge production is expected to be $>9.7 \times 10^7$ tons in 2023 [1]. Harmful components found in sludge, including heavy metals, pathogenic microorganism [2], and various toxic and harmful organic substances, serious impact on human health and living environment [3]. Therefore, it is necessary to pre-treat the sludge before its resourceful disposal or utilization to avoid potential risks of secondary pollution to the environment and human health due to improper treatment [3].

Therefore, sludge need to be treated with appropriate processes, for example: anaerobic digestion, aerobic composting and thermochemical treatment to eliminate the potential for environmental pollution [4]. Anaerobic digestion is an economically and environmentally friendly method that converts organic matter into biogas in the absence of oxygen [5, 6]. However, the resulting high osmotic pressure and inhibition of the microorganisms by high salinity limit the development of anaerobic biological treatment technologies [7, 8]. Aerobic composting process has low energy consumption, low odor generation rate, and can produce organic fertilizer products with certain added value and no harm [9]. Usually supplementary materials or microbial agents are added to the compost pile to optimize the composting effect [10]. However, the purchase of auxiliary materials and microbial agents increases the cost of composting and there are risks in the use of bacterial culture [11]. Although

anaerobic digestion and aerobic composting are the technologies traditionally employed to treat sludge, recent research has found that thermochemical technologies that can further improve energy efficiency [12]. Nowadays, the main technologies for the thermochemical treatment of sludge include gasification, incineration and pyrolysis.

The advantage of gasification disposal technology is that it converts sludge into a mixture of CO, H₂, CH₄ and combustible gaseous hydrocarbons [13]. However, the tar produced during the gasification process can cause many problems with the product gas, such as clogging of downstream piping, valves and injector jets [14].

Incineration disposal technology is used extensively and can completely break pathogens and organic contaminants [15, 16], high degree of non-hazardous, but incineration needs to provide auxiliary fuel and can't achieve the effect of zero waste emissions, so the need to think about flue gas cleaning and fly ash treatment [17].

Compared with sludge gasification and sludge incineration, sludge pyrolysis is the preferred technology due to its high volume reduction rate, low pollutant gas emission, self-sufficiency in energy and high resource recovery rate [18]. Pyrolysis products include H₂, CO and CH₄, which can be used as fuel; tar can be used as liquid fuel and chemical reagent raw material; bio-char has rich pore structure and can be used to make functional materials (soil conditioner, adsorbent, catalyst, etc.) [19].

Sludge pyrolysis technology has the advantages of reduction, non-hazardous and resourcefulness [18] and has become one of the most promising disposal methods for

82 sludge. In addition, pyrolysis technology enables more than 50% of the carbon in
83 dewatered sludge to be sequestered, which also has positive significance in promoting
84 "carbon peaking" and "carbon neutral".

85 Currently the most important sludge pyrolysis technology is sludge catalytic
86 pyrolysis [20, 21]. In sludge catalytic pyrolysis methods, depending on the feedstock
87 and catalyst configuration, they can be divided into in-situ catalysis (where pyrolysis
88 and catalysis occur simultaneously in a single reactor) and ex-situ catalysis (where
89 catalytic reforming of pyrolysis steam (PS) occurs in a second reactor section). Some
90 research has shown that in-situ catalytic pyrolysis may reduce the contact of PS with
91 the catalyst and that coke buildup tends to form on the catalyst surface, leading to
92 problems of reduced catalytic activity [22]. In order to solve the problem, some
93 researchers have suggested that the vapor-catalyst contact reaction should be further
94 optimized in the process of catalysis [22, 23]. However, the tar produced by the
95 catalytic reaction cannot be completely cracked because the catalytic cracking time is
96 too short, leads to the production of heavy tar. And the coalesced tar may polymerize
97 to form more complex compounds, making its removal more difficult [24]. It was
98 assumed that a technology could extend the catalytic time, increase the residence time
99 of PS in the reactor, thereby reducing the complex organic matter content of the coke
100 and increasing the yield of pyrolysis gas (PG) to bio-char. But there are less relevant
101 studies.

102 Therefore, in order to increase the reaction time of sludge pyrolysis, the cyclic

catalytic cracking by pyrolytic steam (CCCPS) process was designed in this study. The main evaluation indexes are pyrolysis product yield, PG fraction, and lower heating value (LHV). In this research process, we debug the different influencing factors such as pyrolysis temperature, iron-based carbon catalyst (IBCC) dosage, and motor speed rate in this process, and also analyzed the nature of sludge pyrolysis products (PG, tar, and bio-char). The mechanism of the CCCPS process was clarified by comparing with ectopic catalytic pyrolysis (ECP) process, and the feasibility of the process was also explored. This process proposes a new sludge resourceful disposal technology based on catalytic reforming of pyrolysis steam cycle. The process realized safe and harmless sludge disposal and efficient resource utilization, providing new insights and reference basis to sludge pyrolysis resourceful disposal technology.

2. Materials and methods

2.1 Sample preparation

Ferric sulfate (≥ 98 wt.%) and other chemical reagents provided by Sigma–Aldrich, USA. The sludge samples used for this experiment were obtained from the municipal residual activated sludge of Xintian Town Wastewater Treatment Plant in Wanzhou, China. The sludge was stabilized by 2 h, The clear liquor was poured off and stored in polyethylene plastic drums and refrigerated at 4 °C to reduce the microbial activity in the sludge for no more than one week in each lot. The sludge was kept in a 20 °C water bath for 30 min before each take to ensure that experimental sludge was in an environment similar to that of the remaining activated sludge from

the wastewater treatment plants. The essential characteristics of the sludge are listed in Table.S1.

2.2 Construction of the experimental setup

The CCCPS process is designed and assembled in-house based on laboratory conditions [25]. This experimental setup is schematically shown in Fig.S1[26], the PS generated by the sludge in the pyrolysis section ⑥ can be circulated in the unit ⑤-⑦ to extend the steam residence time and at the same time increase the number of contacts with the catalyst. The tar and PG enter the tar recovery system ⑧ and the gas collection system ⑩ through polytetrafluoroethylene (PTFE) tubes and their volumes are separately recorded.

2.3 Experimental procedure

In this paper, the sludge was first dewatered using $\text{Fe}_2(\text{SO}_4)_3$ [27], then dried, ground and sieved to 1 - 2 mm, and stored under seal and cold storage for subsequent experiments.

2.3.1 Effect of pyrolysis temperature

Put the quartz boat with 20 g of sludge sample and the empty catalyst carrier into the pyrolysis section area of the machine. The effect of different temperatures (600, 700 and 800 °C) and final pyrolysis temperatures (T600, T700, T800) on sludge pyrolysis was investigated at a motor speed of 1050 r/min. During the experiments, the final temperatures of the catalytic reforming section and the external circulation section were the same as the final temperatures of the pyrolysis section. In this study,

the gas collection system was manually controlled to collect the PG when the final pyrolysis temperature was maintained for 1 h.

2.3.2 Effect of iron-based carbon catalyst dosage

Put the quartz boat with 20 g of sludge sample and the empty catalyst carrier into the pyrolysis section area of the machine. The target temperatures of the pyrolysis section, catalytic reforming section and external circulation section were all set to 800 °C at a motor speed of 1050 r/min, and the effects of different catalyst dosing amounts (0, 4, 8 and 12 g) on the sludge pyrolysis were investigated. The PG collection is similar to that in 2.3.1.

2.3.3 Effect of motor speed

Put the quartz boat with 20 g of sludge sample and 4 g of IBCC and the catalyst carrier into the pyrolysis section area of the machine. The effect of different motor speeds (0, 350, 700, 1050 and 1400 r/min) on the sludge pyrolysis was investigated by adjusting the motor speed to control the gas circulation rate at a fixed target temperature of 800 °C in the pyrolysis section, catalytic reforming section and external circulation section. The PG collection is similar to that in 2.3.1.

2.4 Analysis Methods

In this study, pyrolysis products (bio-char, PG, tar, etc.) were generated from sludge samples using the CCCPS process and the ECP process, respectively, under the same experimental conditions (temperature, IBCC, and motor speed) for comparative analysis.

The mass of bio-char from sludge pyrolysis was obtained by weighting. The mass of PG was obtained by summing the mass of gas collected in each gas bag at different temperatures. And the mass of PG in each gas pocket can be calculated by equation (6). The mass of CD (carbon deposited on the catalyst and catalyst carrier surfaces) was obtained by weighing the difference in mass between the catalyst and carrier before and after catalytic reforming. The mass of coke was obtained by weighing the tar filtrate (the solid remaining on the filter membrane after the liquid phase collected from pyrolysis was filtered through a 0.45 μm PTFE filter membrane). Liquid phase mass was obtained by subtracting the sum of the mass of bio-char, PG, CD and coke from the mass of the sludge sample.

Table.S3 represents the LHV of the gas components (eg. H_2 , CO , CH_4 , C_2H_4 , C_2H_6 , C_3H_6 , C_3H_8 etc.).

2.4.1 Pyrolysis gas analytical methods

PG were analyzed using a gas chromatograph (Agilent 7890B). The PG collected in the gas bags were analyzed with H_2 and N_2 as carrier gases to qualitatively analyze CO , CO_2 , CH_4 , N_2 , C_2H_4 , C_2H_6 , C_3H_6 , C_3H_8 and H_2 , respectively. The H_2 , CO , CO_2 , and N_2 were separated on Agilent J&W HayeSep Q (3 ft, 1/8 in., 2 mm) and Carboxen-1000 (10 ft, 1/8 in., 2 mm) gas-packed columns, respectively, and detected by a Thermal Conductivity Detector (TCD); the C_2H_4 , C_2H_6 , C_3H_6 , and C_3H_8 were separated by HP -PLOT Al_2O_3 S (50 m $\times$ 0.535 mm $\times$ 15 μm) column and detected by the FID detector. Table.S2 shows the instrumental test conditions and warming

procedures. Before each GC test, a calibration equilibrium was preformed using a standard gas mixture, and each gas pocket was tested three times in parallel to ensure the accuracy of the test.

2.4.2 Tar analytical methods

After the liquid phase collected by dichloromethane was filtered through 0.45 μm PTFE membrane, 5 mL of the filtrate was accurately pipetted into a 10 mL centrifuge tube, and 3 g of anhydrous sodium sulfate was added, and then centrifuged with vigorous vibration, 1 mL of the supernatant was extracted into a 1.5 mL liquid-phase vial for analysis, and then the supernatant was analyzed in the liquid-phase vial.

The gas chromatography-mass spectrometry (GC-MS) analysis was carried out on a HP-5 MS column ($60\text{ m} \times 250\text{ }\mu\text{m} \times 0.25\text{ }\mu\text{m}$), and the GC conditions for the GC-MS analysis were as follows: inlet temperature $280\text{ }^{\circ}\text{C}$, gas interface temperature $280\text{ }^{\circ}\text{C}$, carrier gas flow rate 1.5 mL/min ; injection volume $1\text{ }\mu\text{L}$; split ratio 10:1; warming procedure: initial $50\text{ }^{\circ}\text{C}$, hold for 1 min; warming at $5\text{ }^{\circ}\text{C/min}$; and heating up at $5\text{ }^{\circ}\text{C/min}$. 1.5 mL/min ; injection volume $1\text{ }\mu\text{L}$; split ratio 10:1; heating program: initial $50\text{ }^{\circ}\text{C}$, hold 1 min; warm up to $100\text{ }^{\circ}\text{C}$ with $5\text{ }^{\circ}\text{C/min}$, hold 2 min; warm up to $140\text{ }^{\circ}\text{C}$ with $4\text{ }^{\circ}\text{C/min}$, hold 2 min; warm up to $250\text{ }^{\circ}\text{C}$ with $10\text{ }^{\circ}\text{C/min}$, hold 5 min; warm up to $280\text{ }^{\circ}\text{C}$ with $10\text{ }^{\circ}\text{C/min}$, hold 5 min; warm up to $280\text{ }^{\circ}\text{C}$ with $10\text{ }^{\circ}\text{C/min}$, hold 5 min; warm up to $280\text{ }^{\circ}\text{C}$ with $10\text{ }^{\circ}\text{C/min}$, hold 5 min; warm up to $280\text{ }^{\circ}\text{C}$ with $10\text{ }^{\circ}\text{C/min}$, hold 5 min. MS conditions: Ion source temperature $230\text{ }^{\circ}\text{C}$, quadrupole temperature $150\text{ }^{\circ}\text{C}$, EI

ionization 70 eV, full scan 35~550 da. MS data were analyzed and retrieved automatically by the combination of self-built database and NIST17 database.

3.Results and discussion

3.1 Effect of temperature on sludge pyrolysis products

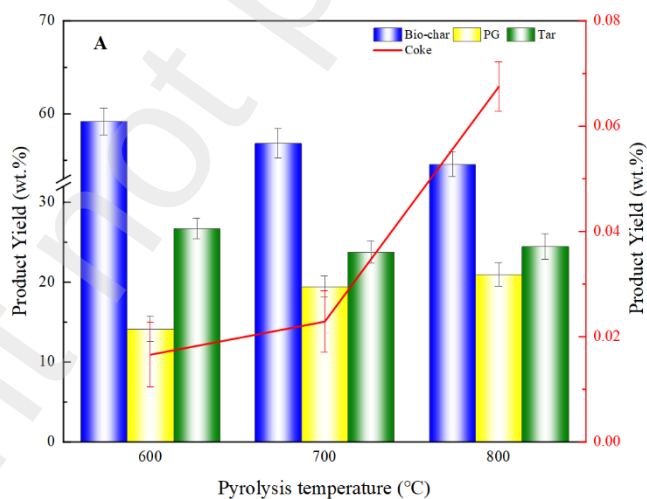
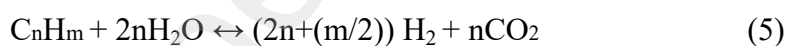
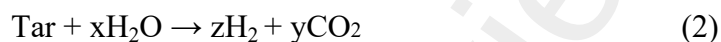
Fig.1A shows the percentage distribution of pyrolysis products at different pyrolysis temperatures. With the increase of pyrolysis temperature, the yield of bio-char and tar showed a decreasing trend, while the yield of PG gradually increased. When the pyrolysis temperature was 800 °C, the bio-char, PG, and tar yields were (54.55 ± 1.34) wt.%, (20.94 ± 1.47) wt.%, and (24.42 ± 1.62) wt.%, respectively. This result shows that at high temperature pyrolysis conditions can promote the production of PG and reduce the yield of bio-char and tar, which is similar to the results of Tao Kan et al [28]. This may be due to the fact that the high temperature promoted the secondary cracking of the tar and enhanced the catalytic reforming reaction of the PS in contact with the bio-char. In addition, the coke yield in the tar increased from (0.016 ± 0.006) wt.% to (0.067 ± 0.004) wt.% under high temperature environment, which may be caused by the partial coking of the tar under high temperature conditions [24, 29].

Fig.1B and 1C show the effect of different temperatures in the pyrolysis and catalytic cycle sections on the distribution of PG components. Fig.1B shows that during the pyrolysis ramping up process, the PG yield increased with increasing pyrolysis temperature when the pyrolysis final temperature was 600 °C, and the PG

yields reached a maximum of (51.12 ± 0.34) wt.% and (37.03 ± 0.54) wt.% at pyrolysis temperatures of 600 °C and 500 °C, respectively, when the pyrolysis final temperature was 700 °C and 800 °C. However, the effect on PG yield was not significant when the pyrolysis section temperature was kept at 800°C with final pyrolysis temperatures of 700 and 800 °C. This shows that the increase in temperature of the catalytic section and the external circulation section (same as the pyrolysis end temperature) promotes the secondary cracking of the tar produced by the sludge pyrolysis and generates more PG [30, 31]. This is supported by the increase in the content of the collected PG components from (12.56 ± 0.42) wt.% to (51.12 ± 0.34) wt.% with the increase in the pyrolysis end temperature when the pyrolysis temperature is the similar 500 °C.

Fig.1C shows the total amount of each component produced by sludge pyrolysis. As the pyrolysis end temperature condition increases, the yield of H_2 increases rapidly from (34.33 ± 5) wt.% to (90.53 ± 7) wt.%, the yield of CO increases from (16.6 ± 1.5) wt.% to (58.25 ± 3.5) wt.%, and CH_4 increases from (19.32 ± 2) wt.% to (45.83 ± 4.5) wt.%. There are two possible reasons for these results. As the temperature of the pyrolysis section increases, more PG is generated by the direct cracking of certain organic substances in the sludge [19]. Based on this CCCPS process, the volatile substances generated by pyrolysis can circulate in the process, which allows these volatile substances to stay longer in the high temperature zone and promotes the re-cracking of these substances by heat to produce H_2 , CO and CH_4 , according to

equations (1) - (5). Therefore, the optimal temperatures for sludge pyrolysis are 500 °C for the starting temperature and 800 °C for the final temperature.



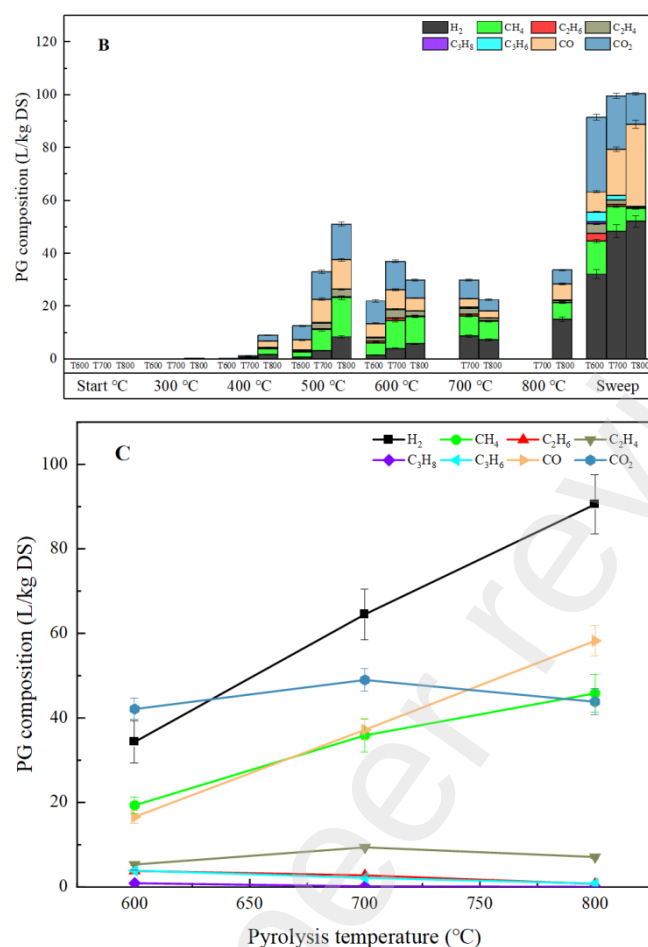


Fig. 1. Effect of temperature on sludge pyrolysis products: **(A)** Effect of temperature on the yield of sludge pyrolysis products; **(B)** Distribution of gas components in each pyrolysis temperature zone; **(C)** Total amount of each component of pyrolysis gas (PG).

3.2 Effect of iron-based carbon catalyst dosing on sludge pyrolysis products

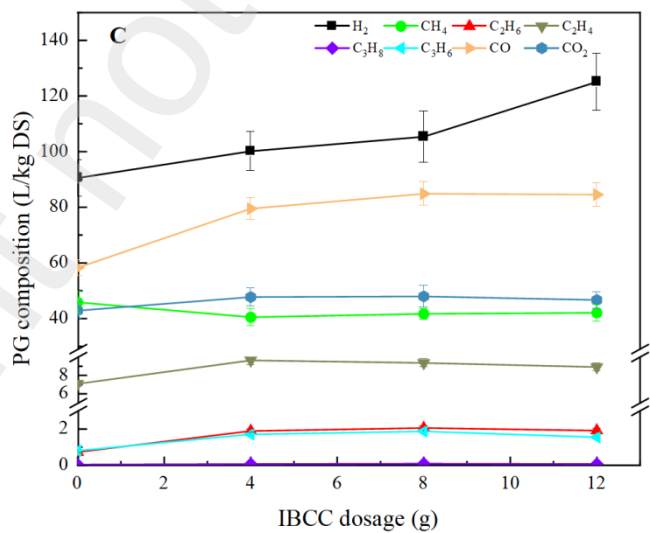
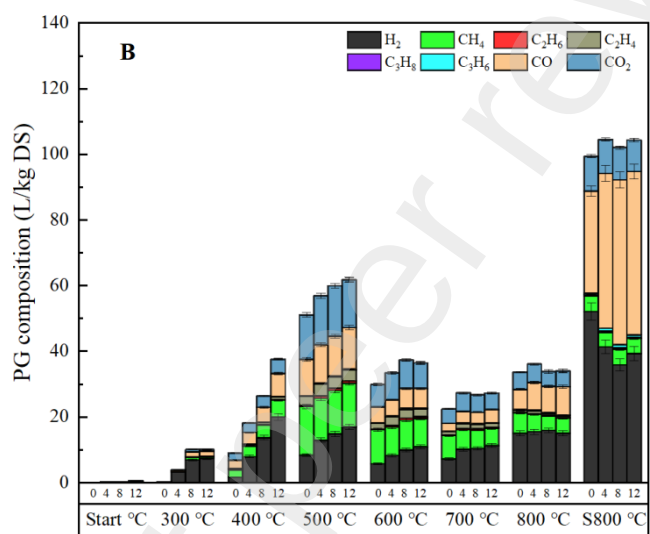
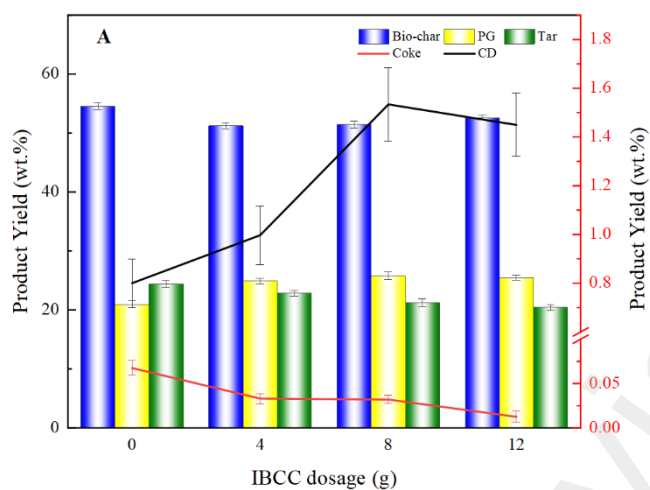
The effect of IBCC dosage on the distribution of sludge pyrolysis product yield is shown in Fig.2A. With the increase of catalyst dosage, the tar yield showed a decreasing trend and decreased from (24.43 ± 0.58) wt.% to (20.44 ± 0.47) wt.% after 12 g of catalyst was added. Meanwhile, the bio-char yield showed a trend of

decreasing and then slightly increasing, from (54.56 ± 0.58) wt.% to (51.22 ± 0.49) wt.% when 4 g of catalyst was added, but increased to (52.60 ± 0.41) wt.% when 12 g of catalyst was added. The PG yield increased with the increase of IBCC addition, and increased from (20.94 ± 0.58) wt.% to (25.01 ± 0.37) wt.% at 4 g of catalyst addition, but further catalyst addition had no significant effect on the improvement of the PG yield. These results indicate that the IBCC have a catalytic effect, allowing some of the tar produced in the pyrolysis section to be catalytically cracked to PG products by the catalyst [32]. The decrease in bio-char yield can be attributed to the added catalyst reducing the deposition of carbon deposited (CD) and tar on the surface of the bio-char and quartz boats during the pyrolysis steam cycle. The increased of CD with the addition of IBCC dosing also explains this. In addition, the coke in the tar also decreased from (0.067 ± 0.008) wt.% to (0.012 ± 0.006) wt.% with the addition of catalyst, which also reduced the difficulty of the tar in subsequent resource utilization.

Fig.2B and 2C show the effect of IBCC dosage on the distribution of PG components. Fig.2B shows that during the warming of the pyrolysis section, the highest yield of PG was collected by the gas bag at 500 °C, indicating that the sludge produced the most PG when the pyrolysis temperature was increased from 400 °C to 500 °C. The maximum PG yield during the whole sludge pyrolysis process was collected at 800 °C (after reaching the final pyrolysis temperature of 800 °C, maintained for 1 h and purged with N₂ 200 mL/min), which can be explained from the viewpoint that during the maintenance of the final pyrolysis temperature, the sludge

continues to thermally explain the PG release until it is completely released; meanwhile, the PS remaining in the unit as a circulating flow is in repeated contact with the bio-char and IBCC, extending the residence time of the PS, which further promotes the catalytic reforming reaction and re-cracking of the PS. From this figure, it can also be found that the PG yield shows an increasing trend with the increase of the IBCC dosing in each pyrolysis temperature section, where the H₂ yield increases particularly. This result is similar to that expressed by [33].

The influence of IBCC dosage on the total amount of all components produced by sludge pyrolysis is shown in Fig.2C. H₂, CO, CH₄ and CO₂ are the main components in PG, when the catalyst dosage increased from 0g to 12g, H₂ rapidly increased from (90.53 ± 6.52) L/kg DS to (125.06 ± 10.25) L/kg DS, accounting for 40.24% of the total PG. CO increased from (79.53 ± 3.97) L/kg DS to (84.53 ± 4.22) L/kg DS, and the yields of C₂H₄, C₂H₆, C₃H₆ and C₃H₈ did not change significantly with catalyst addition and remained at low yields. As shown in Fig. 2D, the LHV of PG increases to (9.59 ± 0.084) MJ/Nm³ with the increase of IBCC dosing and then stabilizes, and the LHV of PG can be regarded as reaching a plateau when the IBCC dosing is 4 g. Therefore, the optimal IBCC dosage is 4 g.



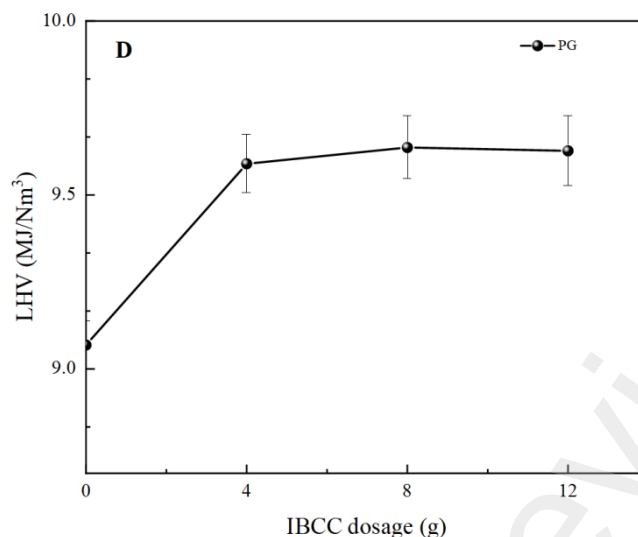


Fig.2. Effect of catalyst dosage on pyrolysis products: **(A)** Effect of iron-based carbon catalyst (IBCC) dosage on the yield of sludge pyrolysis products; **(B)** Distribution of pyrolysis gas (PG) components in each pyrolysis temperature section; **(C)** Total amount of each component of pyrolysis gas (PG); **(D)** Effect of iron-based carbon catalyst (IBCC) dosage on the lower heating value (LHV) of sludge pyrolysis gas (PG).

3.3 Effect of motor speed on sludge pyrolysis products

The speed of motor rotation represents the speed of gas circulation rate in the apparatus. The effect of motor speed on the distribution of sludge pyrolysis product yield is shown in Fig.3A, with the motor speed increasing from 0 r/min to 1400 r/min, the tar yield decreases from (26.33 ± 0.58) wt.% to (19.98 ± 0.56) wt.%, and its yield decreases by 24.12%; the bio-char yield increases slightly from (51.65 ± 0.58) wt. % increased to (53.95 ± 0.54) wt.%; meanwhile, the PG yield showed an increase and then leveled off from (21.02 ± 0.58) wt.% to (24.92 ± 0.62) wt.%, and its yield

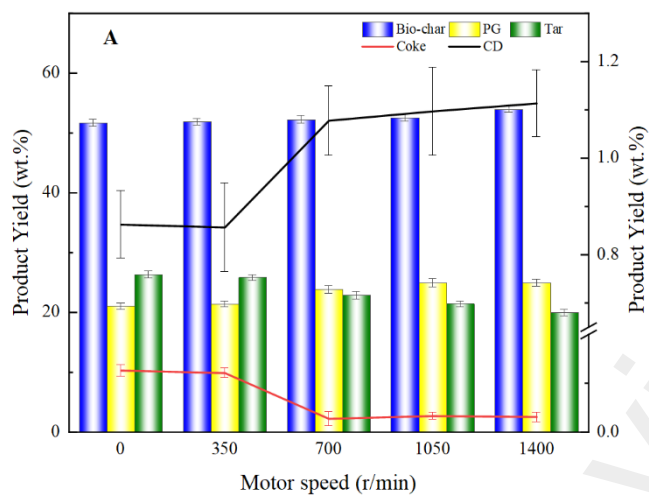
increased by 18.55%. Increasing the motor speed is beneficial to promote sludge pyrolysis, which possibly because higher motor speeds result in an increase in the number of cycles of the pyrolytic stream within the process, thereby increasing the number of times the PS come into contact with the IBCC, thus promoting the catalytic cracking of the tar in the PS and the catalytic reforming with the IBCC, this in turn lead to a slight increase in the bio-char yield and an increase in the CD on the surface of the bio-char and the quartz boat. This idea can also be explained by the increase of CD yield on the surface of the IBCC from (0.86 ± 0.07) wt.% to (1.11 ± 0.06) wt.%. In addition, the coke yield in the tar first decreased to (0.03 ± 0.007) wt.% with increasing motor speed and then leveled off, confirming that increase the number of cycles of the PS flow can reduce the coke yield in the tar and consequently reduce the difficulty of subsequent resource disposal of the tar.

The effect of motor revolutions on the distribution of PG fractions is shown in Fig. 3B and 3C. Fig.3B shows the highest gas production collected by the gas bag at 500 °C during the pyrolysis section ramp-up, which is similar to the results presented in Section 3.2, and indicates that the sludge generates the most PG as the pyrolysis temperature increases from 400 °C to 500 °C. The yield of PG produced at each pyrolysis temperature section showed an increasing trend as the motor revolutions increased. The PG collected at S800 °C has the highest yield in the whole PSCCC process, which is consistent with the findings in section 3.2 shown above. The increase in motor revolution effectively promotes the increase of H₂ components in

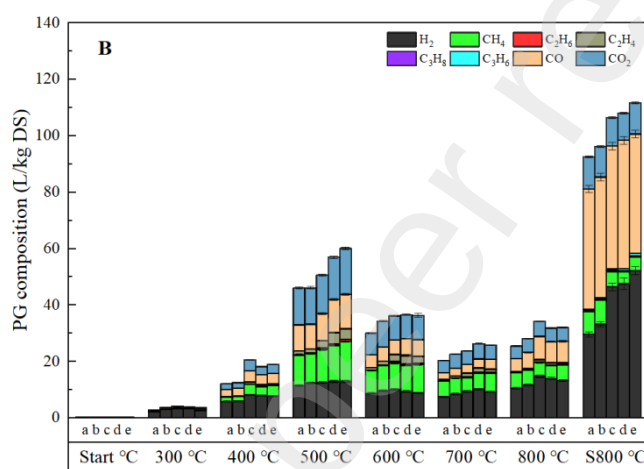
the PG, therefore, it can be interpreted that increasing the PS flow rate promotes the catalytic efficiency of Fe on the surface of IBCC, leading to more H₂ component generation, which is similar to the previous report that Fe has the effect of catalyzing the production of H₂ from sludge pyrolysis [33]. In addition, the variation of motor speed also has different effects on other PG components in each pyrolysis temperature section, for example, the yield of CO₂, CO, and C₂H₄ in PG at 500 °C in the pyrolysis section increases from (13.03 ± 0.39) wt.%, (9.19 ± 0.27) wt.%, (1.33 ± 0.04) wt.% to (16.22 ± 0.49) wt.%, (12.17 ± 0.37) wt.%, (3.61 ± 0.11) wt.%, respectively.

Fig.3C shows the effect of motor speed on the total amount of each component produced by sludge pyrolysis. With the increase of motor rpm, the yields of H₂ and CO increased to (107.18 ± 6) wt.% and (75.31 ± 3.6) wt.%, respectively, with significant changes; the yields of CH₄, CO₂ and C₂H₄ showed a slight increase; the yields of C₂H₆, C₃H₆ and C₃H₈ did not change significantly, thus it can be seen that the effect of motor rpm on the formation of H₂ and CO was the most evident. In addition, as shown in Fig. 3D, the LHV of PG first increases to (9.58 ± 0.15) wt.% with increasing motor speed and then decreases to (9.38 ± 0.15) wt.% but not significantly, therefore, the LHV is reached at a motor speed of 1050 r/min.

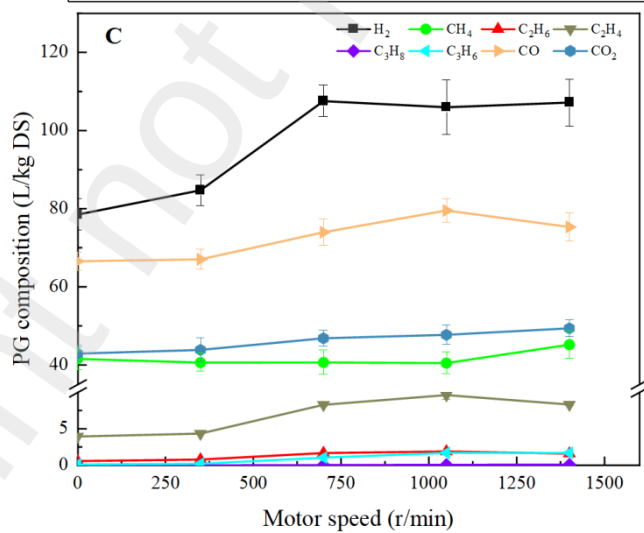
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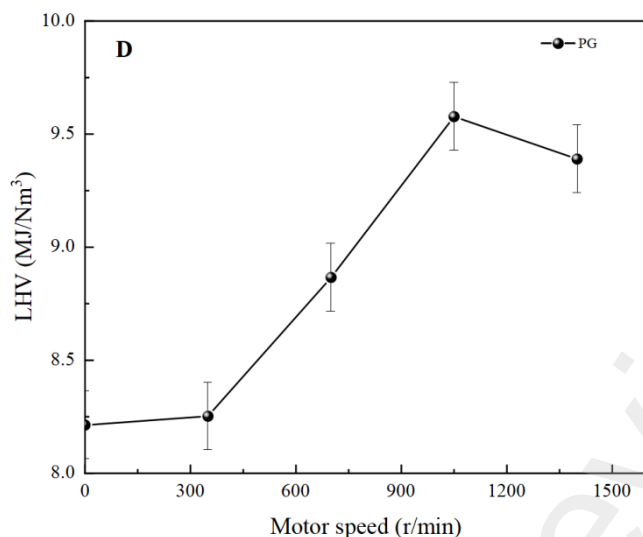


Fig.3. Effect of motor rpm on sludge pyrolysis products: **(A)** Effect of motor rpm on the yield of sludge pyrolysis products; **(B)** Distribution of pyrolysis gas (PG) components in each pyrolysis temperature section; **(C)** Total amount of each component of pyrolysis gas (PG); **(D)** Effect of motor speed on the lower heating value (LHV) of sludge pyrolysis gas PG).

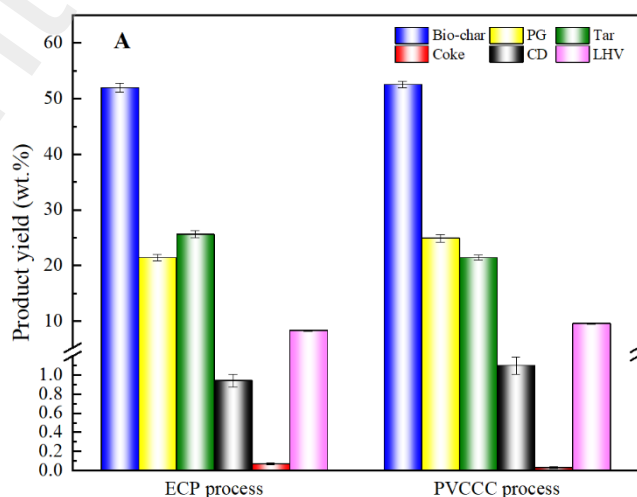
3.4 Mechanism analysis

3.4.1 Analysis of pyrolysis product yields

In order to analyze the contribution of CCCPS process and ECP process to sludge pyrolysis, the comparison of their products was determined in this paper. As shown in Fig.4A, after treatment by the CCCPS process, the tar yield decreased from (25.61 ± 0.64) wt.% to (21.42 ± 0.47) wt.%, the PG production increased slightly from (21.41 ± 0.56) wt.% to (24.91 ± 0.72) wt.%, and the LHV of PG was improved. It is shown that the cycling of PS facilitates the conversion of tar to syngas, which should be due to the fact that CCCPS process increases the number of contacts

between the PS and the IBCC, thus promoting the catalytic cracking of the tar in the PS and the catalytic reforming with IBCC. In addition, the longer residence time of PS from pyrolysis in the CCCPS process contributes to the multiple cracking of tar, thus forming light tar or PG, as shown in a study by Andrés et al [13]. And the CCCPS process also promotes the production rate of bio-char and the formation of CD during the cyclic process, and repeatedly contacted with bio-char and IBCC, which caused some of the tar to be deposited on its surface. This result is well supported by the increase in CD from 0.94 wt.% to 1.1 wt.% shown in Fig.4A.

Fig.4B shows the comparison of the total PG fraction distribution between CCCPS process and ECP process. Unlike the distribution of PG for ECP process, cyclic catalytic pyrolysis increases the yield of H_2 , C_2H_6 , C_2H_4 , C_3H_6 , CO and CO_2 fractions in the PG, which is related to the CCCPS process. In this study, as the CCCPS process can increase the number of contacting steam with IBCC, it may mean that the tar cracking as well as catalytic reforming reactions are promoted and help to increase the fraction yield of PG.



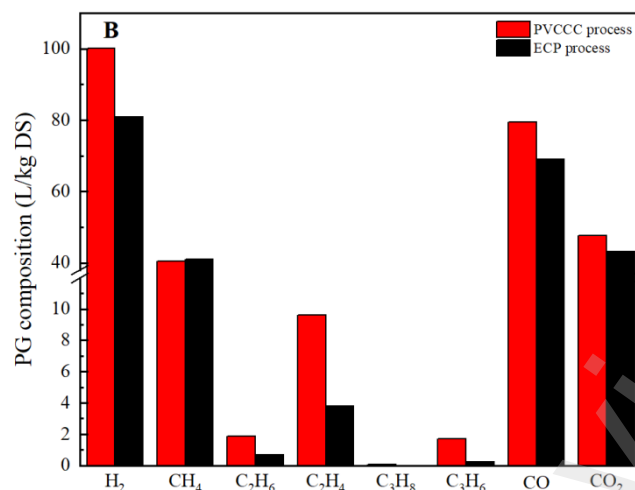


Fig.4. (A) Product yield distribution of the cyclic catalytic cracking by pyrolytic steam (CCCPS) process compared to ectopic catalytic pyrolysis (ECP) process; **(B)** Comparison of pyrolysis gas (PG) composition from the cyclic catalytic cracking by pyrolytic steam (CCCPS) process and ectopic catalytic pyrolysis (ECP) process.

3.4.2 Analysis of tar

According to Table.S4, the organic compounds in sludge tar can be divided into oxygen-containing compounds, nitrogen-containing compounds, hydrocarbons, sulfur-containing compounds, and a few other compounds (including compounds containing elements such as chlorine, bromine, silicon and phosphorus), among which hydrocarbons account for the highest proportion, especially aromatics, which account for 67.94% of the organic phase of tar in CCCPS process and 62.83% in ECP process accounting for 62.83%. It shows that the cyclization of organic matter in the PS is enhanced during the cyclic catalytic pyrolysis of sludge. This may be caused by the prolonged PS residence time in the CCCPS process, which is similar to the study by

Andres et al [13]. More types of oxygenated organic compounds (OOCs), including alcohols, aldehydes, esters, carboxylic acids, and furans, were detected in the organic phase of tar of the ECP process, which accounted for 10.52% of the total organic phase, whereas OOCs of esters, carboxylic acids, and furans were detected in the organic phase of the tar subjected to CCCPS process, which accounted for 1.38% of the organic phase, which showed that the percentage of the organic content was greatly reduced. Moreover, the cyclic catalytic pyrolysis generated more CO and CO₂, indicating that this process promotes the re-decomposition of the OOCs into CO and CO₂, which further reduced the OOC in the organic phase of the tar effectively. It is reported that OOCs are associated with certain unfavorable properties of tars: non-volatility, high viscosity, corrosiveness, poor calorific value, thermal instability and immiscibility with fossil fuels [34]. Therefore, the tar produced in CCCPS process is more conducive to subsequent resource utilization than ECP process. In this study, different types of nitrogenous compounds including nitriles, pyrroles, pyridines, indoles, quinolines, phenanthridines and carbazoles were also detected in the tar phase [35]. Tar organic phase accounted for 22.68% and 21.77% in CCCPS and ECP process, respectively. This may imply that the proteins of nitrogenous compounds in sludge were broken down and cyclized during the pyrolysis process, which is similar to the previous study by Chen et al [36], that CCCPS process also promotes the cyclization process of nitrogenous organic compounds. It was also found that the organic phase of the tar contained compounds of other elements and small amounts of

sulfur-containing compounds, mainly in the form of thiophene sulfur.

Fig.5 shows the GC-MS chromatograms of the organic phase in the tar produced by the CCCPS process and the ECP process. Unlike the CCCPS process, it was found that in the organic phase of tar of the ECP process, there existed a substance with a longer peak time than that of the organic phase of tar of the CCCPS, while some peaks of the organic phase of tar of the CCCPS process became higher at short residence times, indicating the presence of organic matter with higher molecular weight in the organic phase of tar of ECP process, and there were more organics with smaller molecular weights in the organic phase of tar of CCCPS. This result proves that the large molecular organics in tar will be cleaved into small molecular organics and PG in the CCCPS process.

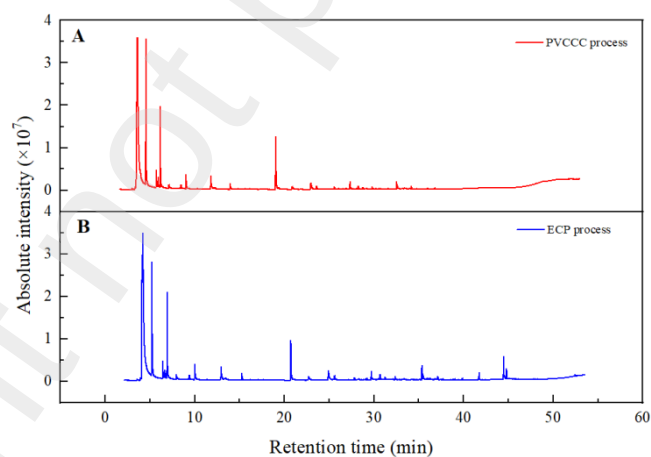


Fig.5. Comparison of GC-MS chromatogram of organic phase in tar from cyclic catalytic cracking by pyrolytic steam (CCCPS) process and ectopic catalytic pyrolysis (ECP) process.

3.4.3 Potential Mechanisms

As shown in Fig.6, in the ECP process the residence time of the PS in the

pyrolysis process is short and the tar produced is heavy tar. In the CCCPS process, there may be an underlying mechanism whereby the residence time of the PS in the recirculation process is extended, repeatedly coming into contact with the bio-char and IBCC, so that the heavy tar undergoes a secondary cracking process, partly generating a gas and partly condensing to the main component of light tar. Fig.6 likewise expresses the process of secondary cracking of tar, where the oxygenated organic matter in heavy tar decomposes into CO and CO₂, and the nitrogenous organic matter decomposes into pyrroles, pyridines, and indoles small molecule organic matter [36]. Fig.5 also proves that the organic phase of tar in the cyclic catalytic unit has more small molecular weight organics, so the CCCPS process is feasible for solving the problem of large molecular organics in heavy tar that are difficult to be resourced again.

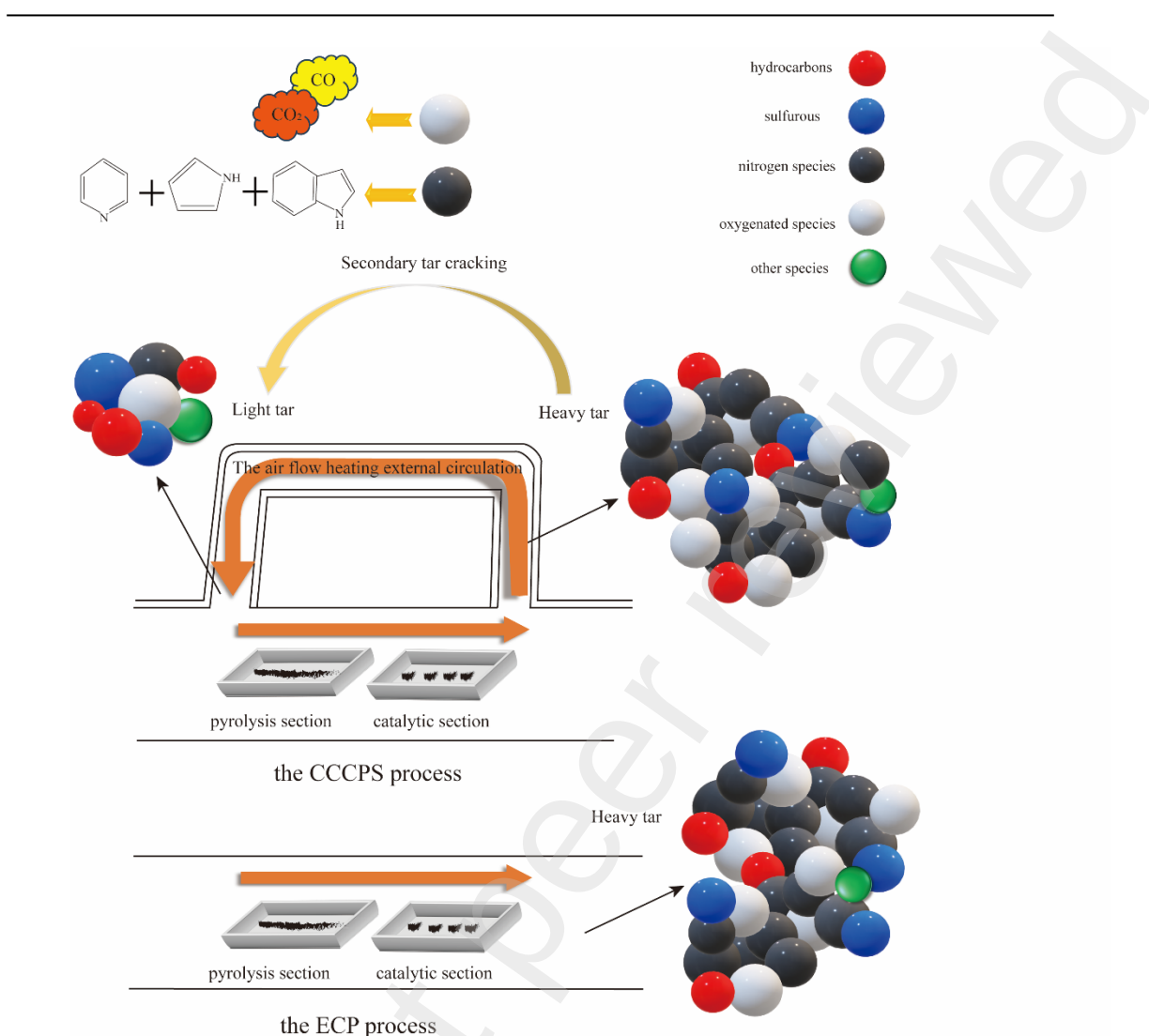


Fig.6. Comparison of tar cracking processes in cyclic catalytic cracking by pyrolytic steam (CCCPS) process and ectopic catalytic pyrolysis (ECP) process.

4. Conclusions

Under the optimal conditions (pyrolysis starting temperature was 500 °C, final temperature was 800 °C, IBCC dosage was 4 g, motor speed was 1050 r/min), the PG yield was (24.92 ± 0.62) wt.%, the tar yield was (19.98 ± 0.56) wt.%, the bio-char yield was (53.95 ± 0.54) wt.%. Compared with the ECP process, the PG yield increased by 3.5 wt.%, the tar yield decreased by 4.19 wt.%, the CD increased to 0.16

wt.%, and the LHV of the PG was improved. The PS become more contact with IBCC and bio-char, and the tar's large organic molecules are cracked again.

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Supporting Information

Supporting information for this work can be found in the online electronic version of this paper.

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